Animal Image Classification Using Deep Learning

A Deep Learning Approach to Classify 15 Animal Species

Akash A

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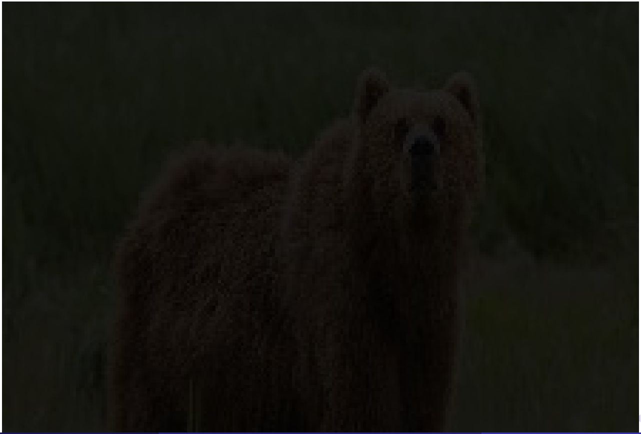
Project Overview

- **Objective:** Build an AI system to classify animals in images into 15 different classes
- **Dataset:** 15 animal categories (Bear, Bird, Cat, Cow, Deer, etc.)
- **Approach:** Transfer Learning with EfficientNetB0
- **Key Features:**
  - Data Augmentation for better generalization
  - Model fine-tuning for improved accuracy
  - Performance evaluation using validation metrics

Dataset Description

- **Total Classes:** 15
- **Image Dimensions:** 224x224x3 (RGB)
- **Dataset Structure:**
  - Each animal has a separate folder (e.g., `animals/bear/`, `animals/cat/`)



Methodology

1. **Data Preprocessing**
   - Image Augmentation (Rotation, flipping, zooming)
   - Train-Validation Split (80% Training, 20% Validation)
2. **Model Architecture**
   - Base Model: EfficientNetB0 (pre-trained on ImageNet)
   - Custom Layers: Dense (1024 neurons), Dropout (0.5)
3. **Training**
   - Optimizer: Adam (lr=0.0001)
   - Loss: Categorical Cross-Entropy
   - Callbacks: Early Stopping & Model Checkpointing

Model Training & Performance

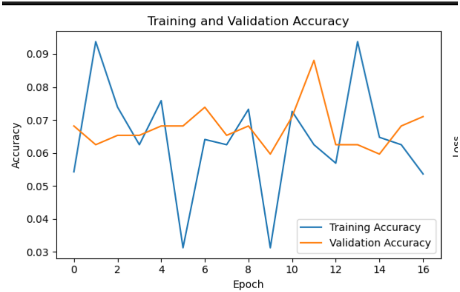


Figure: Training vs Validation Accuracy

Figure: Training vs Validation Loss

Challenges & Solutions

Challenge	Solution
Limited training data per class	Used Transfer Learning (EfficientNetB0)
Overfitting risk	Added Dropout and Data Augmentation
Class imbalance	Ensured stratified sampling in train-val split

Future Improvements

- Expand Dataset: Include more animal classes
- Hyperparameter Tuning: Optimize learning rate, batch size
- Deployment: Convert model to TensorFlow Lite for mobile use
- Real-time Classification: Integrate with a webcam application

Conclusion

- Successfully built an animal classifier with high accuracy
- Demonstrated the effectiveness of Transfer Learning
- Potential applications in:
  - Wildlife monitoring
  - Pet recognition systems
  - Zoo management solutions

Thank You!

Any Questions?